



Kalibrierschein

Calibration Certificate

Gegenstand: Photodiode
Object:

Hersteller: Hamamatsu / ANL
Manufacturer:

Typ: enclosed S3584-06
Type:

Kennnummer: DP 000429
Serial No.:

Auftraggeber: Argonne National Laboratory
Applicant: 9700 S. Cass Ave
Lemont IL 60439
USA

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Reference No.:

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Calibration mark:

Ort der Kalibrierung: PTB Berlin
Location of calibration:

Datum der Kalibrierung: 2024-02-01
Date of calibration:

Im Auftrag Berlin, 2024-03-13
On behalf of PTB



Dr. Christian Gollwitzer

Siegel
Seal



Im Auftrag
On behalf of PTB



Levent Cibik

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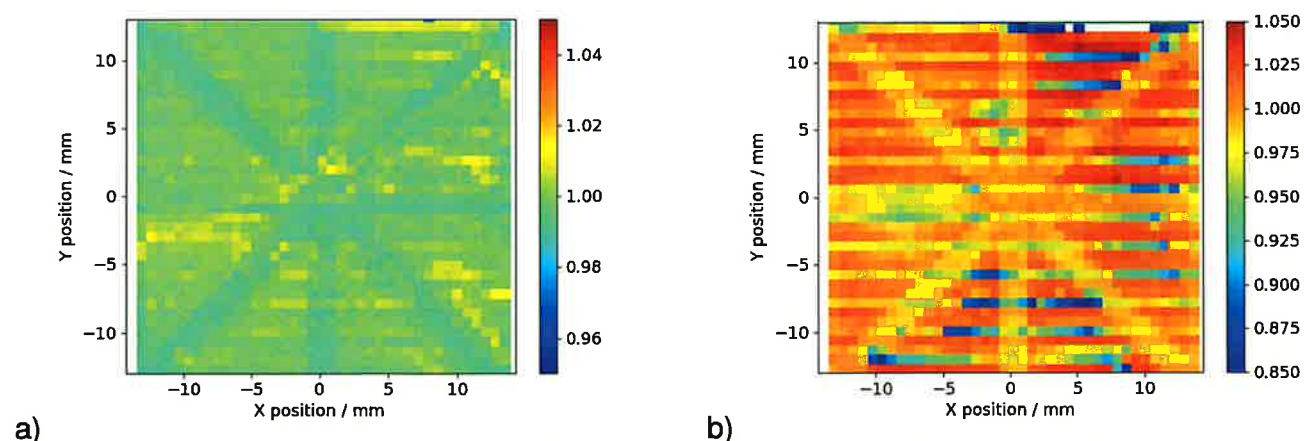


Figure 1: Homogeneity of the spectral responsivity at 20 keV (a) and 27.5 keV (b). The responsivity was mapped with a beam size of approximately 0.7 mm × 0.7 mm and normalized to the centre of the detector.

Description of photodiode

The radiation detector is a silicon photo diode with an area of 28 mm × 28 mm enclosed in a metal housing with a filter. The detector was electrically connected in such a way that a positive photocurrent was measured. No bias voltage was applied. No sensitivity of the detector to polarization was assumed.

Description of the measuring station

The measurements were performed at the BAMline, a wavelength-shifter beamline [1] at the electron storage ring BESSY II. A double multilayer monochromator (DMM) and a double crystal monochromator (DCM) were used in series up to 27.5 keV to combine a high spectral resolution and higher order contributions below 10^{-4} . At higher energies, the higher order contributions are below that level by using only the DCM. The photocurrents of the diodes were measured with calibrated picoammeters (Keithley 617). The beam was below $1.3 \times 1.3 \text{ mm}^2$ in size for the calibration. To account for variations of the stored electron current, the photocurrents were normalized to the current of an ion chamber operated in transmission in the monochromatic beam.

The monochromatized synchrotron radiation is linearly polarized to better than 90 %. The radiant power used for the calibration measurements ranged between 0.49 μW and 5.9 μW .

Sample adjustment

The centre of the detector under test is adjusted relative to the incoming radiation using linear translation stages. A scan across the active area provides information on the spatial homogeneity of the detector. As shown in Figure 1, spatial variations are mainly below 1 % at 20 keV and below 5 % at 27.5 keV. The larger inhomogeneity at 27.5 keV may be due to silver fluorescence [2].

Measurement

The responsivity of the diode is determined by comparison to a reference detector of known responsivity. Here, a reference photodiode directly calibrated by comparison to the PTB primary detector standard,

a cryogenic electrical substitution radiometer [3, 2] was used as reference detector. For the calibration, the radiant power in a monochromatic beam is measured at selected photon energies with the reference detector. Second, the photocurrent of the detector under test is measured in its centre at the same photon energies. At the end of a measurement cycle the radiant power is remeasured with the reference detector to detect any possible instabilities caused e.g. by instabilities of the electron beam position.

Result of the calibration

Table 1: Spectral responsivity of the detector and measurement uncertainty.

Photon Energy /eV	Spectral Responsivity /A W ⁻¹	Uncertainty ($k = 2$) /A W ⁻¹
8000	0.2637	0.0022
9000	0.2560	0.0021
10 000	0.2421	0.0020
12 500	0.1882	0.0016
15 000	0.1354	0.0011
17 500	0.095 49	0.000 86
20 000	0.068 13	0.000 62
22 500	0.049 59	0.000 40
25 000	0.036 83	0.000 48
27 500	0.035 86	0.000 29
30 000	0.027 96	0.000 23
35 000	0.017 62	0.000 15
40 000	0.011 80	0.000 10
45 000	0.008 26	0.000 07
50 000	0.006 01	0.000 07
55 000	0.004 51	0.000 06
60 000	0.003 50	0.000 05

The measurement results are represented in Table 1. The uncertainty stated is the expanded measurement uncertainty obtained by multiplying the standard measurement uncertainty by the coverage factor $k = 2$. It has been determined in accordance with the "Guide to the Expression of Uncertainty in Measurement (GUM)". The value of the measurand then normally lies, with a probability of approximately 95 %, within the attributed coverage interval. The measurement uncertainty encompasses contributions from the cryogenic electrical substitution radiometer, possible photocurrent variations during the calibration, the dark current of the detector, the picoammeter calibration, radiation of a higher order and scattered radiation in the monochromatized beam. The relative uncertainty of the photon energy scale of the monochromator is below 0.1 %. The uncertainties given in Table 1 represent only the uncertainty of the measured responsivity and do not account for the inhomogeneity of the detector or instability under irradiation. The measured values are valid only for the radiant power range mentioned above and for the time of calibration.

References

- [1] W Görner, MP Hentschel, BR Müller, Heinrich Riesemeier, M Krumrey, G Ulm, W Diete, U Klein, and R Frahm. BAMline: the first hard X-ray beamline at BESSY II. *Nucl. Instrum. Meth. A*, 467: 703–706, 2001.
- [2] M Gerlach, M Krumrey, L Cibik, P Müller, H Rabus, and G Ulm. Cryogenic radiometry in the hard X-ray range. *Metrologia*, 45:577, 2008.
- [3] H Rabus, V Persch, and G Ulm. Synchrotron-radiation-operated cryogenic electrical-substitution radiometer as the high-accuracy primary detector standard in the ultraviolet, vacuum-ultraviolet, and soft X-ray spectral ranges. *Applied optics*, 36:5421–5440, 1997.

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